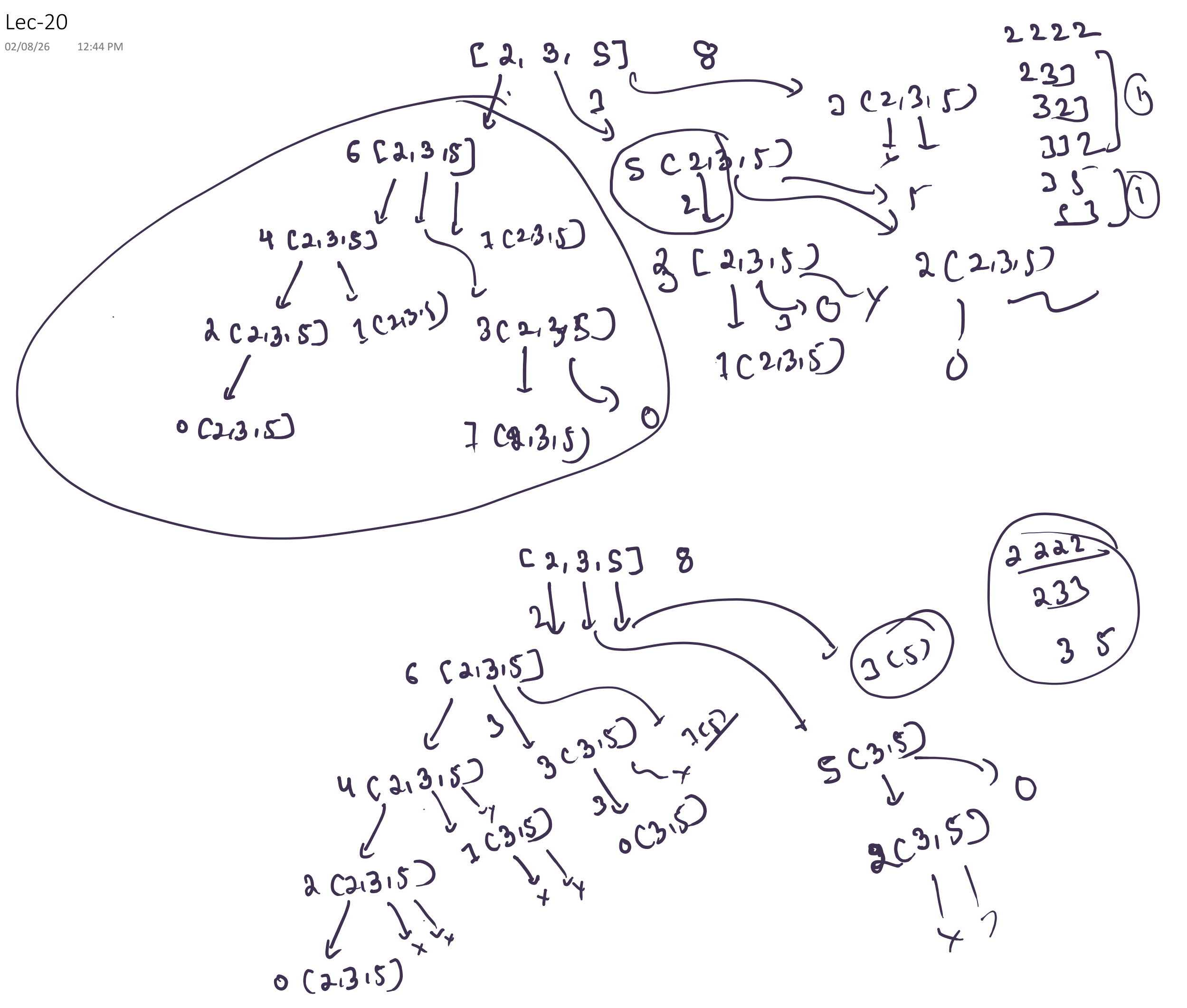
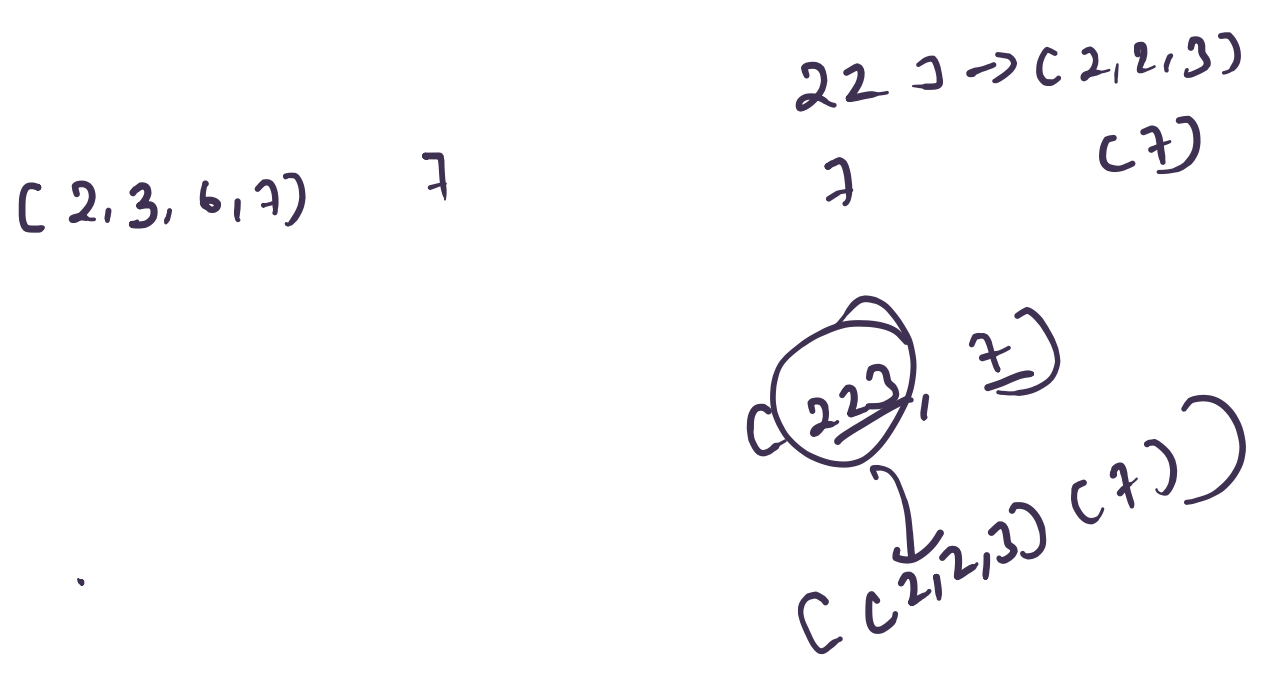
Lec-20

02/08/26 12:44 PM

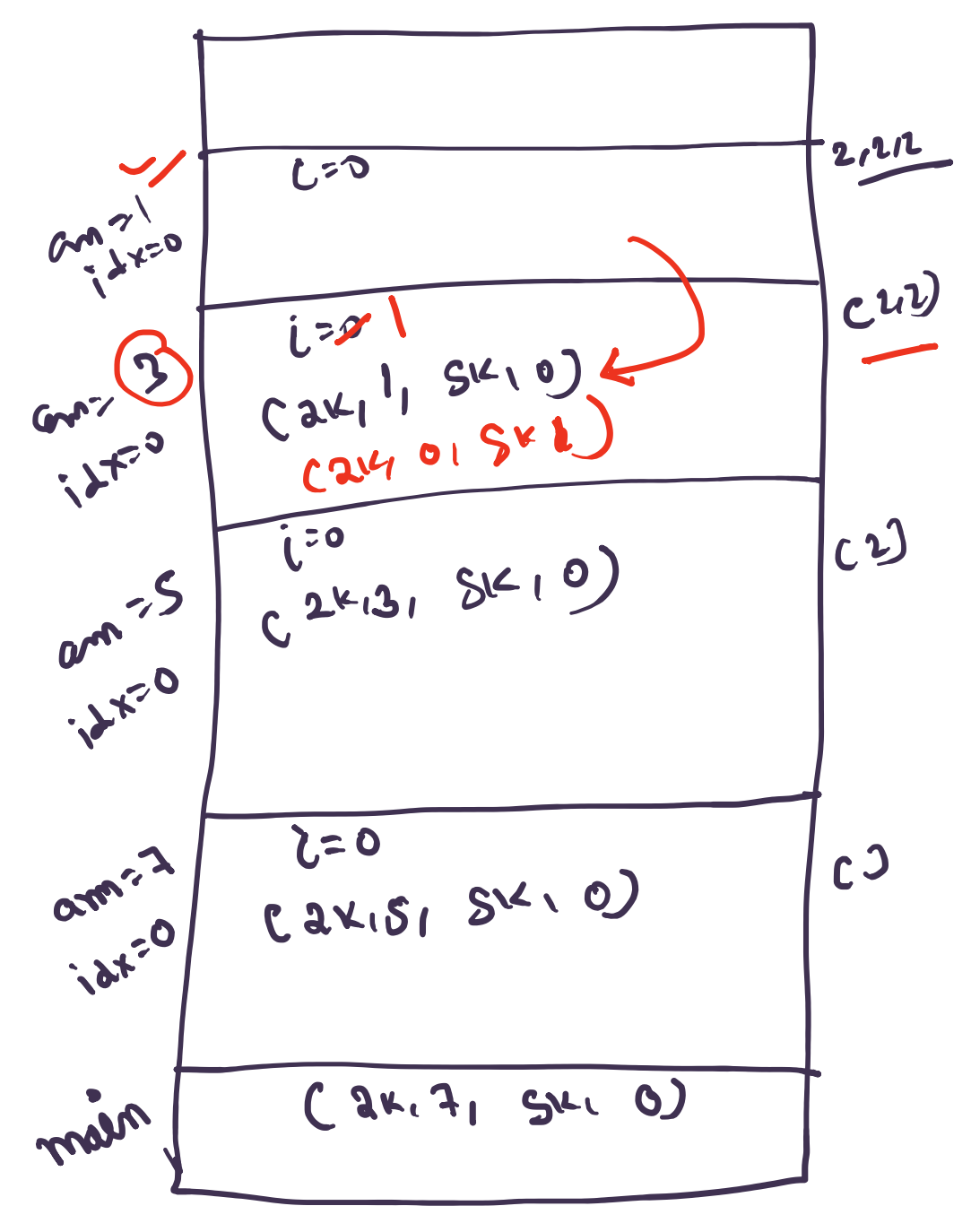

[2, 3, 5] 8
6 [2,3,5]
4 [2,3,5]
1 [2,3,5]
2 [2,3,5]
1 [2,3,5]
3 [2,3,5]
0 [2,3,5]
1 [2,3,5]
5 [2,3,5]
3 [2,3,5]
2 [2,3,5]
2222
233
323
332
35
53
6 [2,3,5]
4 [2,3,5]
3 [3,5]
2 [2,3,5]
1 [3,5]
0 [3,5]
0 [2,3,5]
5 [3,5]
2 [3,5]
3 (5)
2222
233
3 5


[2, 3, 6, 7] 7
22 3 -> [2,2,3]
7 [7]
(223, 7)
[[2,2,3] [7]]

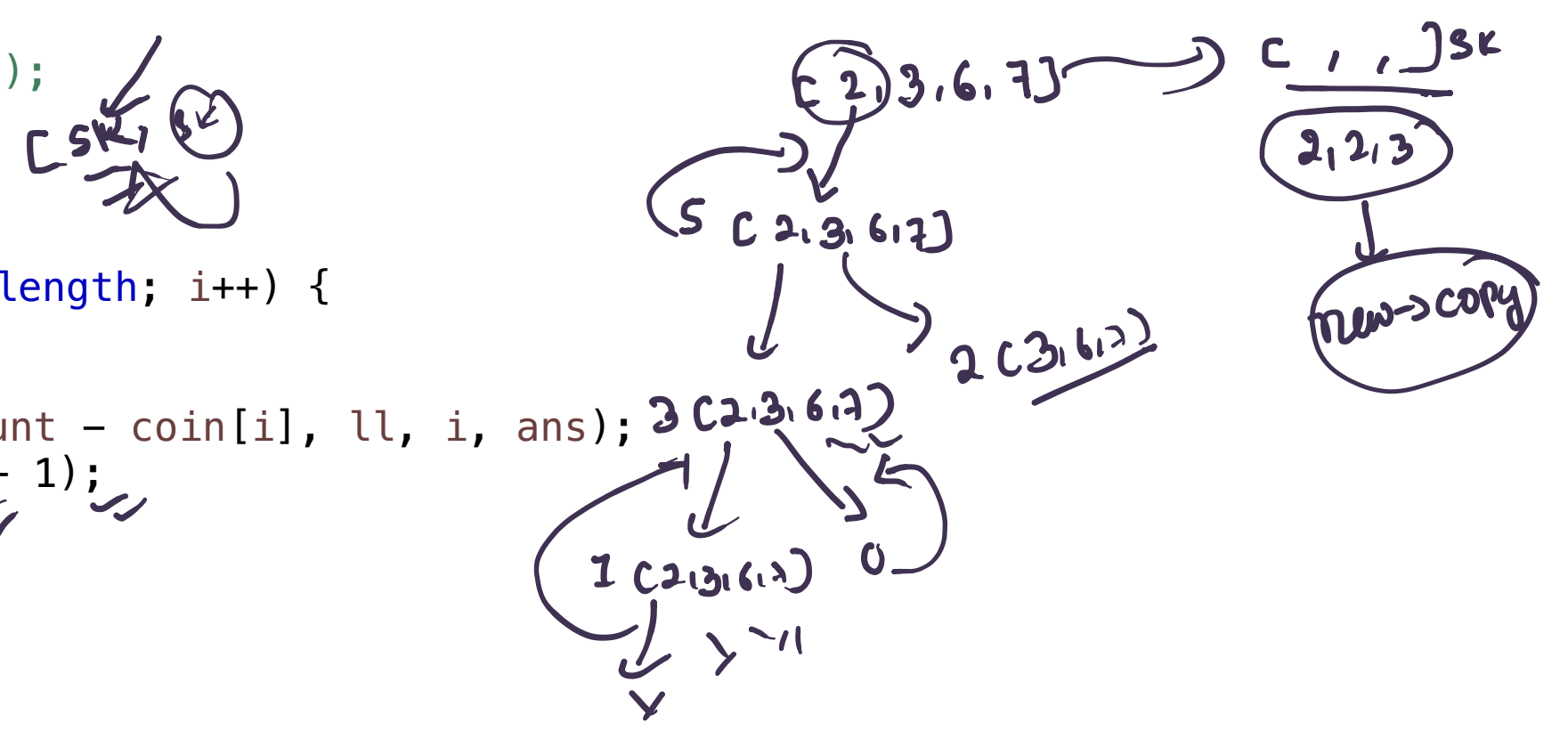
```
public static void Combinaton(int[] coin, int amount, List<Integer>ll,int idx) {
        if(amount==0) {
            System.out.println(ll);
            return;
        }

        for (int i = idx; i < coin.length; i++) {
            if(amount>=coin[i]) {
                ll.add(coin[i]);
                Combinaton(coin,amount-coin[i],ll,i);
            }
        }

    }
```

[2,3,6,7]

[2,2,2,3]

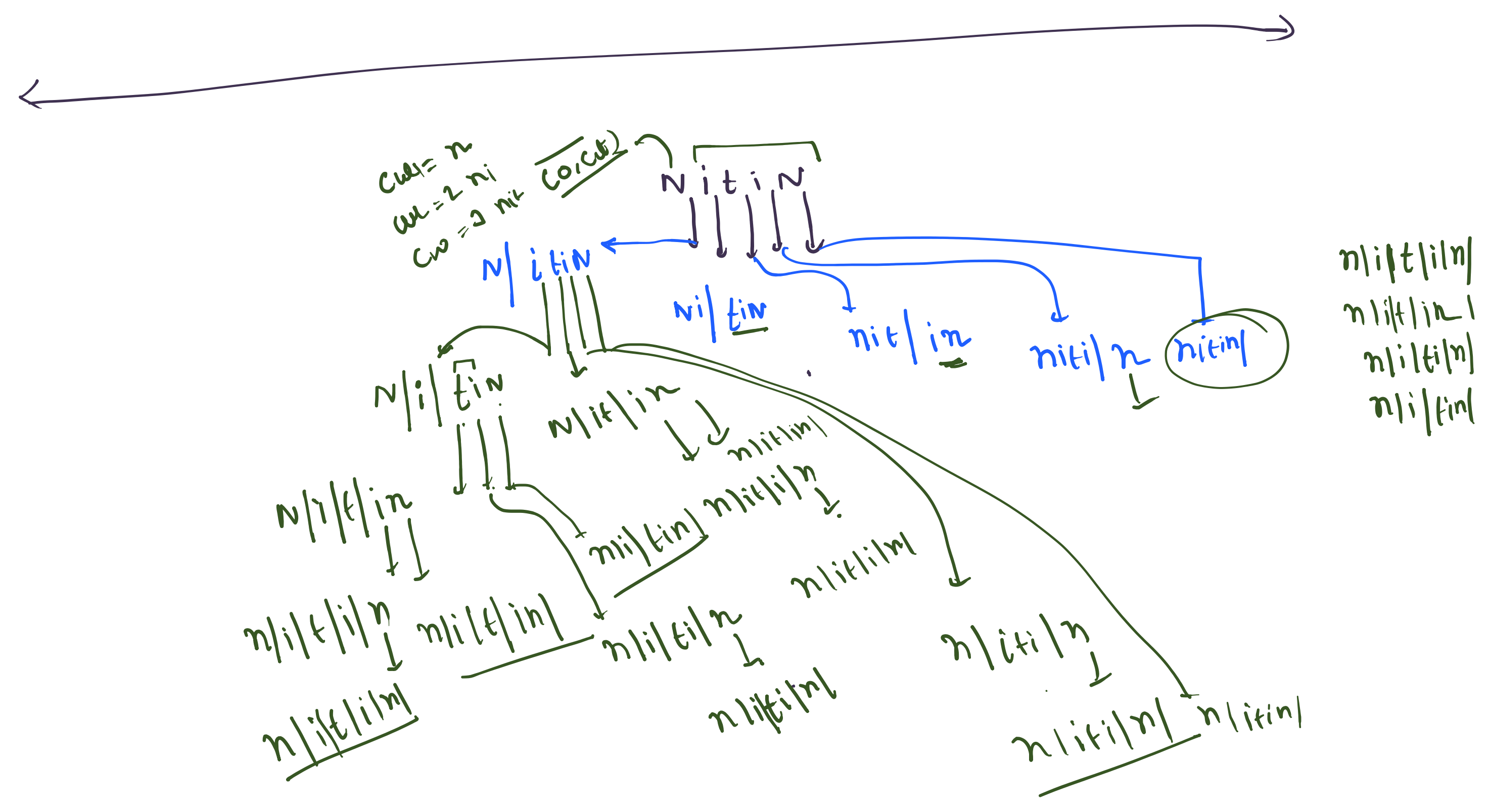

am=1 idx=0
i=0
2,2,2
am=3 idx=0
i=0 1
(2k, 1, 5k, 0)
(2k, 0, 5k, 1)
[2,2]
am=5 idx=0
i=0
(2k, 3, 5k, 0)
[2]
am=7 idx=0
i=0
(2k, 5, 5k, 0)
[]
main
(2k, 7, 5k, 0)

```
public static void Combinaton(int[] coin, int amount, List<Integer> ll, int idx, List<List<Integer>> ans) {
        if (amount == 0) {
            // System.out.println(ll);
            ans.add(ll);
            return;
        }

        for (int i = idx; i < coin.length; i++) {
            if (amount >= coin[i]) {
                ll.add(coin[i]);
                Combinaton(coin, amount - coin[i], ll, i, ans);
                ll.remove(ll.size() - 1);
            }
        }

    }
```


[5k, 8k]
[2, 3, 6, 7]
[, ,] 5k
2,2,3
new->copy
5 [2,3,6,7]
2 [3,6,7]
3 [2,3,6,7]
1 [2,3,6,7]
0

(n,i,t,i,n)
(n,i,ti,n)
(n,iti,n)
nitin
n|i|t|i|n|
n|iti|n
nitin|

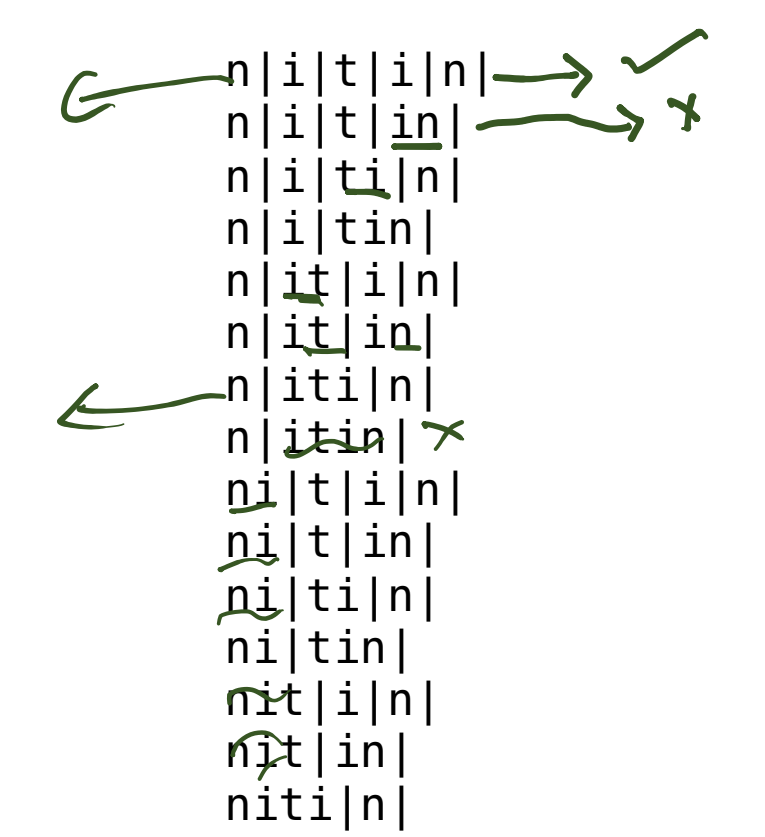
```
n|i|t|i|n|
n|i|t|in|
n|i|ti|n|
n|i|tin|
n|it|i|n|
n|it|in|
n|iti|n|
n|itin|
ni|t|i|n|
ni|t|in|
ni|ti|n|
ni|tin|
nit|i|n|
nit|in|
niti|n|
nitin|
```

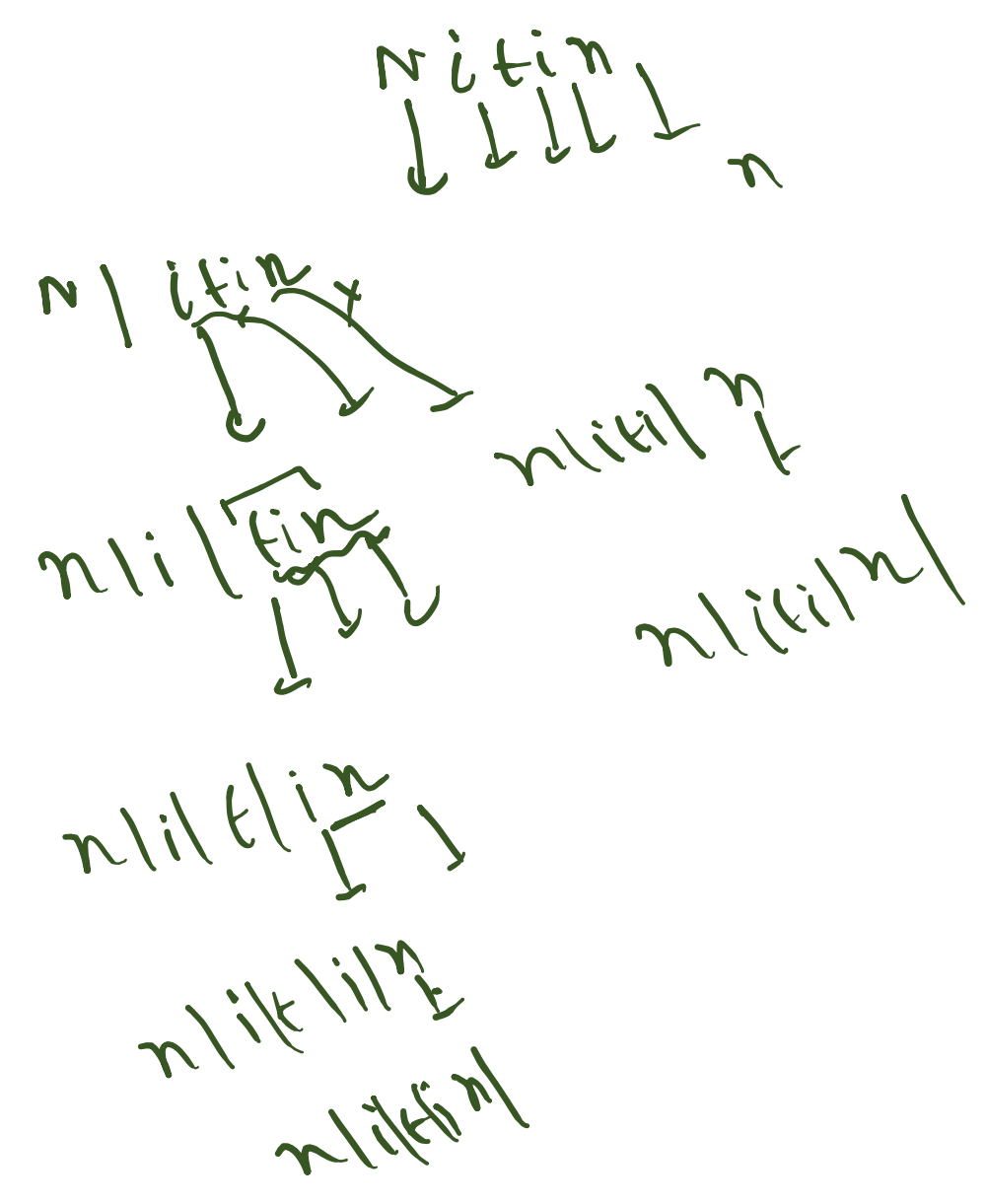

cut=n
cut=2 ni
cut=3 nit
(0,cut)
N i t i N
N| itiN
Ni| tiN
nit| in
niti| n
nitin|
N|i| tiN
N|it|in
N|i|t|in
n|i|t|i|n
n|i|t|in|
n|i|ti|n
n|i|t|i|n|
n|it|in|
n|iti|n
n|iti|n|
n|itin|
n|i|t|i|n|
n|i|t|in|
n|i|ti|n|
n|i|tin|

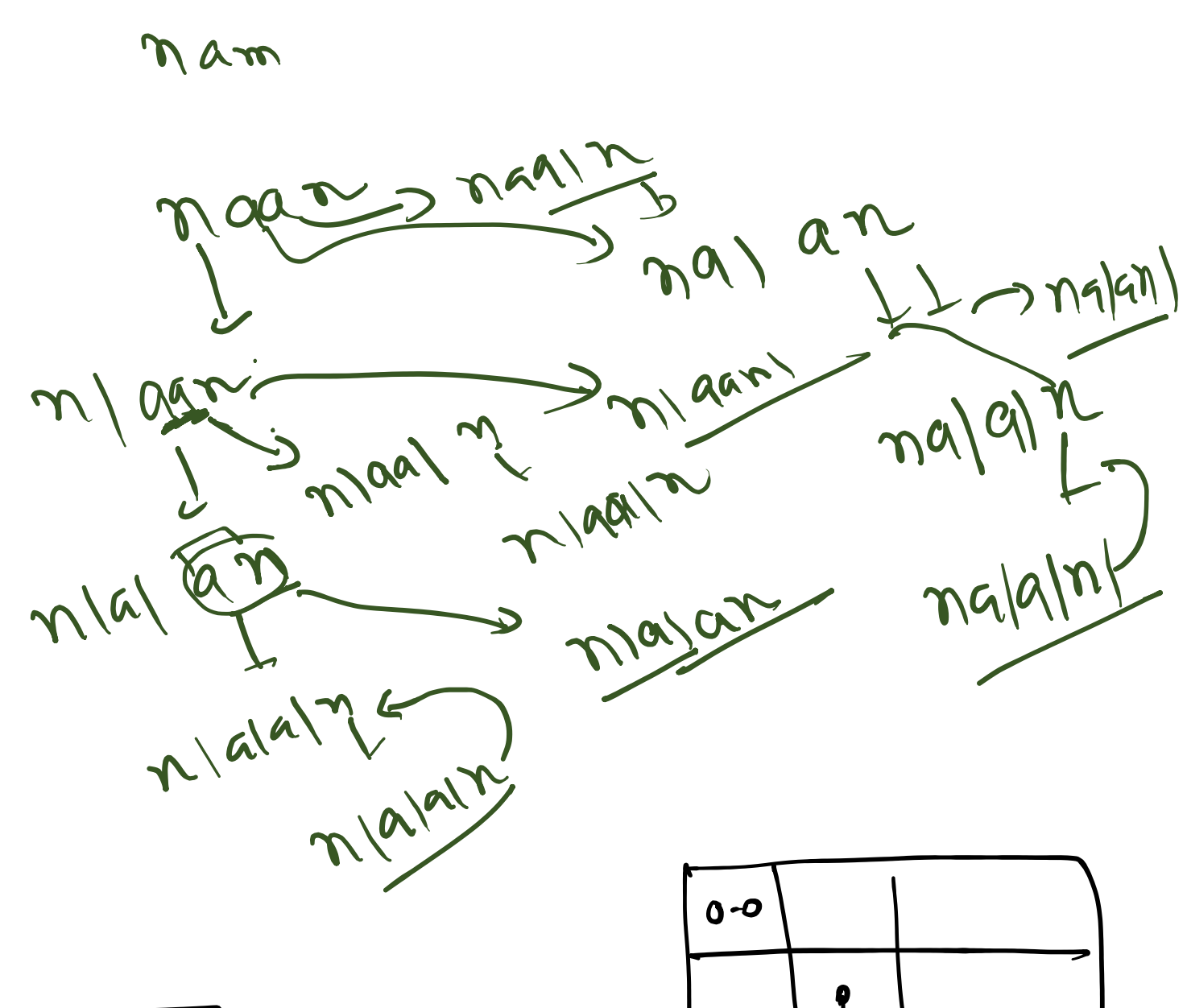
N i ti N

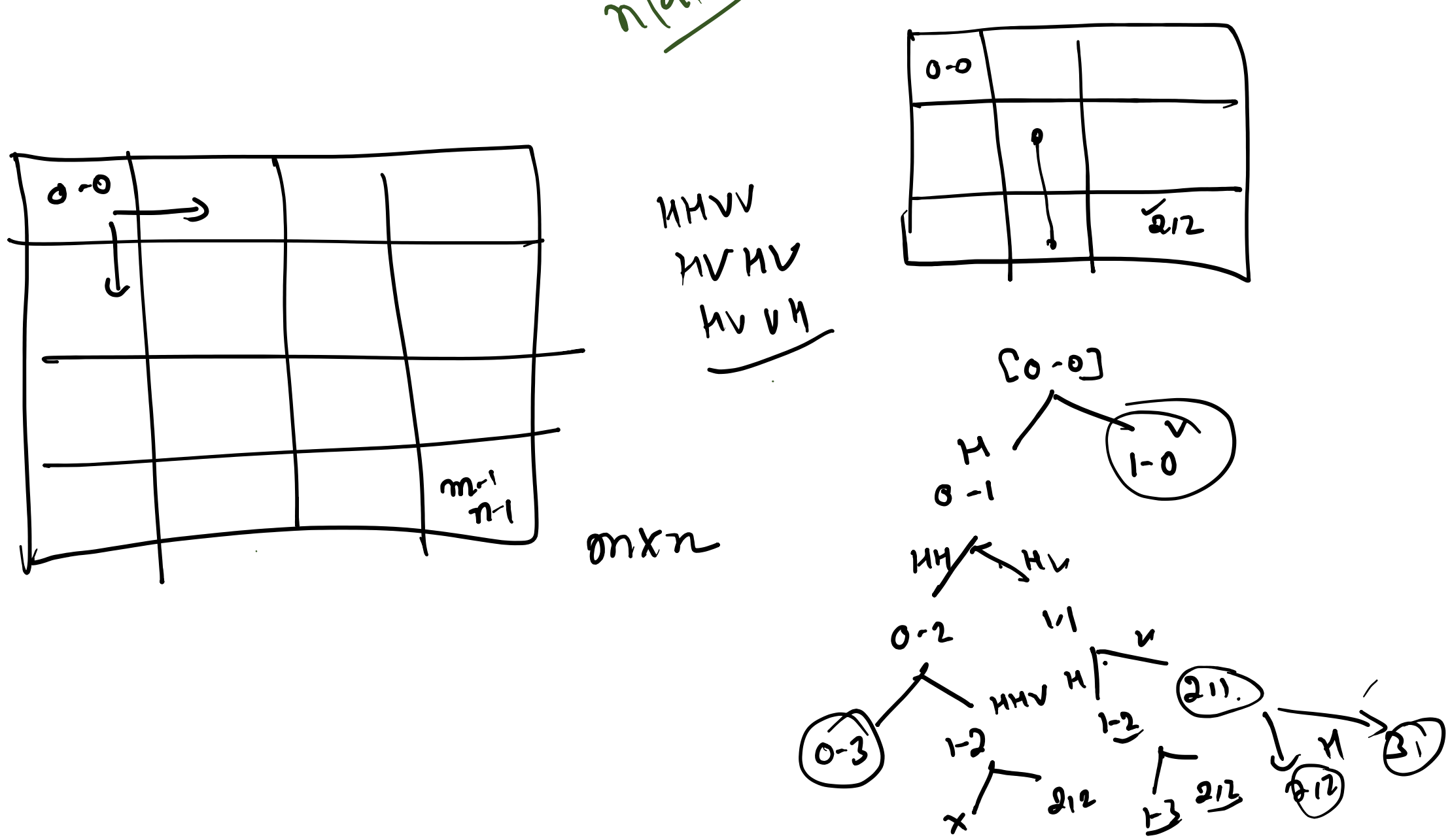
cut=1	itin (1)	n - (0,1)
cut=2	tin (2)	ni (0,2)
cut=3	in (3)	nit (0,3)
cut=4	n (4)	nitin (0,4)
cut=5	— (5) ub	niti n (0,5)

```
n|i|t|i|n|
n|i|t|in|
n|i|ti|n|
n|i|tin|
n|it|i|n|
n|it|in|
n|iti|n|
n|itin|
ni|t|i|n|
ni|t|in|
ni|ti|n|
ni|tin|
nit|i|n|
nit|in|
niti|n|
nitin|
```




nitin
n
n| itin
n|i| tin
n|iti| n
n|iti|n|
n|i|t| in
n|i|t|i|n
n|i|t|i|n|


nan
naan
naa|n
na| an
na|a|n|
n| aan
n|aa| n
n|aan|
na|a| n
na|a|n|
n|a| an
n|a|a| n
n|a|a|n
n|a|an


0-0
m-1
n-1
mxn
HHVV
HVHV
HVVH
0-0
2,2
[0-0]
H
V
1-0
0-1
HH
HV
0-2
1,1
0-3
HHV
1-2
2,1
3,1
2,2